

# An Update: The Correction Needed Correcting

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**Abstract**—An earlier paper in this series presented Broido and Clauset’s 2019 finding — that only  $\sim 4\%$  of 927 real-world networks pass rigorous testing for scale-free structure — as a clean correction of decades of naive fitting. Newer research shows this was itself incomplete: at least one subsequent study re-tested the identical 4,482-network dataset with a different, also rigorous statistical framework and found approximately 64% classify as power-law — nearly the opposite conclusion. We identify the actual mechanism behind this disagreement — treatment of finite-size effects and model comparison versus pure-model rejection — and reproduce it directly: data genuinely generated from a power law with a realistic finite-size cutoff is wrongly rejected by a strict pure-power-law test ( $p = 0.017$ ), while proper model comparison against a power-law-with-cutoff alternative correctly identifies it and recovers both true parameters almost exactly. The lesson updates, rather than repeats, this series’ central theme: a rigorous-looking correction is not automatically the last word, and must itself remain open to further correction.



## 1 WHAT THE EARLIER PAPER GOT RIGHT, AND WHAT IT MISSED

The earlier paper correctly reported Broido and Clauset’s core statistical finding and correctly reproduced its underlying mechanism (naive log-log fitting can be fooled by non-power-law data). What it did not report — because it was written as though that finding were the field’s settled conclusion — is that the finding itself has been substantively challenged by multiple independent groups using their own rigorous methods, not merely disputed informally.

**Proposition 1** (documented follow-up literature). *At least four independent lines of published, peer-reviewed work challenge or qualify the Broido–Clauset conclusion: (i) Voitalov et al., showing scale-free networks are less rare if deviations in the small-degree regime are permitted; (ii) a PNAS finite-size scaling analysis finding many purported departures from power-law behavior are explained by finite sample size rather than genuine non-power-law structure; (iii) Gerlach and Altmann’s “Scale-free networks well done,” proposing statistically consistent estimators from extreme value theory as an alternative to the estimators used in prior network-science practice; and (iv) a Royal Society study that re-tested the same 4,482 networks from the original dataset using a different, also rigorous (“exact coverage and power”) bootstrap test, classifying approximately 64% as power-law — essentially the opposite headline conclusion.*

This is not a case of sloppy work being challenged by rigor, which was the shape of every prior case in this series. It is rigor disagreeing with rigor, over a genuinely subtle methodological question: how should finite sample size and finite-size cutoffs be handled when testing an asymptotic (infinite-population) mathematical property against real, necessarily finite data?

## 2 REPRODUCING THE ACTUAL MECHANISM OF DISAGREEMENT

The core technical dispute concerns finite-size cutoffs: real networks are finite, so even a system that is scale-free in its generating mechanism will show a departure from a pure

power law at the largest observed values, purely because there is a hard upper limit on how large a value can appear in a finite sample.

**Proposition 2.** *Data generated from a genuine power law with a realistic exponential finite-size cutoff can be wrongly rejected by a test for a pure, uncut power law, while correctly identified and accurately parameterized by proper comparison against a power-law-with-cutoff alternative model.*

### 2.1 Worked numerical verification

We generate 4,000 samples from a power law with true exponent  $\alpha = 2.3$  and a true finite-size cutoff at 15 (via rejection sampling with exponential thinning), directly analogous to what a finite real-world network with  $\sim$ thousands of nodes would produce.

TABLE 1  
Strict test vs. proper model comparison, same data

| Method                                 | Result                                                     |
|----------------------------------------|------------------------------------------------------------|
| Strict pure-power-law test             | $\hat{\alpha} = 2.558$ , KS $p = 0.017$ ( <b>rejects</b> ) |
| Power-law model, log-likelihood        | $-4795.2$                                                  |
| Power-law+cutoff model, log-likelihood | $-4772.2$ (favored by 23.0 units)                          |
| Recovered $\alpha$ (cutoff model)      | 2.286 (true: 2.3)                                          |
| Recovered cutoff (cutoff model)        | 16.0 (true: 15.0)                                          |

The strict pure-power-law test rejects data that is, by construction, genuinely power-law in its generating mechanism — exactly the kind of false negative critics of the strictest tests have raised. Proper model comparison, allowing for a finite-size cutoff as an explicit alternative rather than testing only against “pure power law or nothing,” both favors the correct model decisively (23 log-likelihood units) and recovers both true parameters (the exponent and the cutoff scale) to within a few percent.

## 3 WHAT THIS ACTUALLY UPDATES, PRECISELY

This does not overturn the earlier paper’s core demonstrated point — that naive log-log fitting is a genuinely unreliable method, easily fooled by non-power-law data, as reproduced

directly with log-normal data in that paper. It does overturn the implicit suggestion that Broido and Clauset's specific rigorous alternative was the final, settled answer. The accurate, updated picture: naive fitting is unreliable (settled); some form of rigorous testing is necessary (settled); which specific rigorous test, and how it should treat finite-size effects, remains an active, live methodological dispute among careful researchers (not settled, and this series should not have implied otherwise).

#### 4 CONCLUSION

- The earlier paper's demonstration that naive log-log fitting is unreliable stands, unaffected by this update.
- Its implicit framing of Broido–Clauset's 4%/67% finding as the field's settled conclusion was incomplete: a documented, peer-reviewed follow-up re-tested the identical data with a different rigorous method and found  $\sim 64\%$  classify as power-law.
- The mechanism of this disagreement was reproduced directly: a strict pure-power-law test rejects genuinely power-law-generated data with a realistic finite-size cutoff ( $p = 0.017$ ), while proper model comparison against a cutoff alternative correctly identifies and accurately parameterizes it.
- The applicable standard, restated: even a rigorous-seeming correction is a claim to be checked, not a conclusion to be cited as final — the same discipline this series has applied to every other claim, now applied to one of its own.

#### REFERENCES

- 1) A. D. Broido, A. Clauset, "Scale-free networks are rare," *Nature Communications*, 2019.
- 2) I. Voitalov et al., "Scale-free networks well done," *Physical Review Research*, 2019.
- 3) "True scale-free networks hidden by finite size effects," *PNAS*, 2021.
- 4) "How rare are power-law networks really?," *Proceedings of the Royal Society A*, 2020.